

Drug Class	Specific Drugs	Mechanism of Action	Spectrum of Activity	Resistance Mechanisms	Adverse Effects	Notes
Beta-lactams	Penicillin G, Cephalosporins (e.g., Cefazolin), Carbapenems (Imipenem), Monobactams (Aztreonam)	Inhibit bacterial cell wall synthesis by binding to PBPs and blocking transpeptidation.	Gram-positive bacteria, Gram-negative cocci, non-beta-lactamase-producing anaerobes.	Beta-lactamase enzymes, altered PBPs, reduced permeability, efflux pumps.	Hypersensitivity (e.g., rash, anaphylaxis), nephritis, diarrhea.	Include subcategories: natural penicillins, beta-lactamase-resistant penicillins, extended-spectrum penicillins, and antipseudomonal penicillins.
Glycopeptides	Vancomycin, Telavancin	Inhibit cell wall synthesis by binding to D-Ala-D-Ala of peptidoglycan precursors; damages membranes.	Gram-positive bacteria, including MRSA and beta-lactam-resistant organisms.	Altered target site (D-Ala-D-Lac substitution), reduced permeability.	Red man syndrome, nephrotoxicity, ototoxicity, chills, fever.	Used for severe Gram-positive infections (e.g., MRSA, C. difficile colitis).
Aminoglycosides	Gentamicin, Amikacin, Streptomycin, Neomycin	Inhibit bacterial protein synthesis by binding irreversibly to the 30S ribosomal subunit.	Aerobic Gram-negative bacteria; limited activity against Gram-positive bacteria.	Enzymatic inactivation, altered ribosomal binding site, reduced uptake.	Nephrotoxicity, ototoxicity, respiratory paralysis at high doses.	Bactericidal; often combined with beta-lactams for synergy. Not effective against anaerobes.
Tetracyclines	Doxycycline, Minocycline, Tigecycline	Bind reversibly to the 30S ribosomal subunit, blocking aminoacyl-tRNA binding.	Broad-spectrum: Gram-positive, Gram-negative, Rickettsiae, Chlamydia, Mycoplasma.	Efflux pumps, ribosomal protection proteins, enzymatic inactivation.	GI upset, photosensitivity, enamel discoloration in children, hepatotoxicity.	Contraindicated in pregnant women and children under 8 years. Tigecycline has activity against multidrug-resistant pathogens.
Macrolides	Erythromycin, Azithromycin, Clarithromycin	Bind to 50S ribosomal subunit, preventing peptide chain elongation.	Gram-positive cocci, atypical pathogens (e.g., Mycoplasma).	Efflux pumps, ribosomal methylation, enzymatic degradation.	GI upset, QT prolongation, drug interactions (CYP450 inhibition).	Azithromycin requires less frequent dosing and has fewer interactions.

			Chlamydia).			
Sulfonamides	Sulfamethoxazole, Trimethoprim (Co-trimoxazole)	Inhibit folate synthesis by competing with PABA and blocking dihydrofolate reductase.	Broad-spectrum: Gram-positive and Gram-negative bacteria, Nocardia, Chlamydia.	Overproduction of PABA, altered enzyme affinity, reduced drug uptake.	Crystalluria, hypersensitivity, Stevens-Johnson syndrome, kernicterus in neonates.	Often combined with trimethoprim for synergistic effects; first-line for Pneumocystis jirovecii pneumonia.
Fluoroquinolones	Ciprofloxacin, Levofloxacin, Moxifloxacin	Inhibit DNA gyrase (topoisomerase II) and topoisomerase IV, preventing DNA replication.	Broad-spectrum with Gram-negative, Gram-positive, and atypical activity.	Altered DNA gyrase/topoisomerase, efflux pumps.	GI upset, CNS effects, cartilage damage (avoid in children), QT prolongation.	Used for UTIs, respiratory infections, traveler's diarrhea; avoid in pregnancy.
Miscellaneous	Metronidazole, Polymyxins (e.g., Colistin)	Metronidazole damages DNA; Polymyxins disrupt cell membranes.	Anaerobes (e.g., Clostridium difficile) for metronidazole; Gram-negative rods for polymyxins.	Rare for metronidazole; polymyxins: resistance due to altered target sites.	Peripheral neuropathy, nephrotoxicity (polymyxins).	Metronidazole has disulfiram-like effects with alcohol. Polymyxins are restricted to resistant infections.

Comprehensive Table of Antibacterial Agents